Gavin's Lectures on Physics - Quantum Mechanics



Author：长颈鹿 Gavin

October 28, 2026

Contents

1 Foundational Postulates & Early Quantum Tools **3**
- 1.1 Fundamental Postulates . . . 3
  - 1.1.1 State Postulate (Wave Function) . . . 3
  - 1.1.2 Born Interpretation . . . 3
  - 1.1.3 Correspondence Postulate (Operators) . . . 3
  - 1.1.4 Measurement Postulate (Eigenvalue Equation) . . . 3
  - 1.1.5 Collapse Postulate . . . 3
  - 1.1.6 Evolution Postulate (Schrödinger Equation) . . . 3
  - 1.1.7 Spin-Statistics Postulate . . . 3
- 1.2 Essential Formulas . . . 4
  - 1.2.1 Origin of Operators . . . 4
  - 1.2.2 The Schrödinger Equation . . . 4
  - 1.2.3 Commutation Relations . . . 4
  - 1.2.4 Heisenberg Uncertainty Principle . . . 4
  - 1.2.5 Representation Transformation . . . 5
- 1.3 Ehrenfest Theorem . . . 5

2 1D Motion **6**
- 2.1 The 1D Infinite Square Well . . . 6
- 2.2 The 1D Harmonic Oscillator . . . 6

3 3D Motion, Angular Momentum & Spin **8**
- 3.1 Two-Body Problem . . . 8
- 3.2 Angular Momentum and Spin . . . 8
- 3.3 Stern–Gerlach Experiment . . . 9
- 3.4 Molecular Rotational Energy Levels . . . 9
- 3.5 Clebsch–Gordan Coefficients . . . 10

4 Hydrogen Atom & Stationary Perturbation Theory **11**
- 4.1 Spherically Symmetric Potential and the Hydrogen Atom . . . 11
  - 4.1.1 General radial equation . . . 11
  - 4.1.2 Angular solutions . . . 11
  - 4.1.3 Hydrogen atom –zeroth order . . . 11
- 4.2 Time-Independent Perturbation Theory . . . 13
  - 4.2.1 Non-degenerate case . . . 13
  - 4.2.2 Degenerate case . . . 13
- 4.3 Fine Structure of the Hydrogen Atom . . . 14
  - 4.3.1 Relativistic correction . . . 14
  - 4.3.2 Spin-orbit interaction(S-L) . . . 14
  - 4.3.3 Lamb shift . . . 16
- 4.4 Hyperfine Structure of the Hydrogen Atom (I-S) . . . 16

5 Atoms in External Magnetic Fields **18**
- 5.1 Electron Spin Resonance (B-S) . . . 18
- 5.2 Nuclear Magnetic Resonance (B-I) . . . 18
- 5.3 Normal Zeeman effect (B-L) . . . 18
- 5.4 Anomalous Zeeman effect (weak B - L+S coupling) . . . 18

5.5 Paschen–Back effect (strong B - L+S decoupled) 19

6 Many-Body Systems & Identical Particles 20
6.1 Identical Particles 20
6.2 Helium Atom 20
6.3 Bose–Einstein Condensation (BEC) 21

7 TIME-DEPENDENT PERTURBATION & TRANSITIONS 22
7.1 Interaction Picture 22
7.2 Dyson Series 22
7.3 Transition Probability 22
7.4 Constant Perturbation 23
7.5 Fermi's Golden Rule 24
7.6 Harmonic Perturbation 24
7.7 Rabi Oscillation 25
7.8 Electron transition for H atom 25
7.9 Selection Rules for Electric Dipole (E1) Transitions 26

1 Foundational Postulates & Early Quantum Tools

1.1 Fundamental Postulates

1.1.1 State Postulate (Wave Function)

The state of a physical system is completely described by a wave function $\Psi(\mathbf{r}, t)$.

1.1.2 Born Interpretation

$$P(\mathbf{r}, t) = |\Psi(\mathbf{r}, t)|^2, \qquad \int_{-\infty}^{\infty} |\Psi(\mathbf{r}, t)|^2 \, d^3r = 1. \tag{1}$$

1.1.3 Correspondence Postulate (Operators)

Every measurable physical quantity corresponds to a linear, Hermitian operator.

$$\hat{x} = x, \tag{2}$$
$$\hat{p} = -i\hbar\nabla, \tag{3}$$
$$\hat{T} = \frac{\hat{p}^2}{2m} = -\frac{\hbar^2}{2m}\nabla^2, \tag{4}$$
$$\hat{H} = \hat{T} + V(\mathbf{x}) = -\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{x}). \tag{5}$$

1.1.4 Measurement Postulate (Eigenvalue Equation)

$$\hat{A}\Psi = a\Psi, \tag{6}$$

where Ψ is an eigenfunction of $\hat{A}$ with eigenvalue a.

1.1.5 Collapse Postulate

Immediately after a measurement yielding eigenvalue a_n, the state collapses instantaneously to the corresponding eigenfunction Ψ_n.

1.1.6 Evolution Postulate (Schrödinger Equation)

$$i\hbar\frac{\partial}{\partial t}\Psi(\mathbf{r}, t) = \hat{H}\Psi(\mathbf{r}, t). \tag{7}$$

The wave function of a closed system evolves deterministically according to the time-dependent Schrödinger equation.

1.1.7 Spin-Statistics Postulate

- **Fermions**: half-integer spin, anti-symmetric total wave function, obey Pauli exclusion principle.
- **Bosons**: integer spin, symmetric total wave function.

1.2 Essential Formulas

1.2.1 Origin of Operators

For a matter wave $\Psi = \Psi_0 e^{i(kx-\omega t)}$.

de Broglie wave tells momentum: $p = \hbar k = \frac{h}{\lambda}$:

$$-i\hbar\frac{\partial\Psi}{\partial x} = -i\hbar(ik)\Psi = \hbar k\Psi = p\Psi, \tag{8}$$

hence

$$\hat{p} = -i\hbar\frac{\partial}{\partial x} \xrightarrow{3D} \hat{p} = -i\hbar\nabla. \tag{9}$$

Einstein photon tells energy: $E = \hbar\omega$

$$i\hbar\frac{\partial\Psi}{\partial t} = E\Psi = \hat{H}\Psi, \tag{10}$$

hence

$$\hat{H} = i\hbar\frac{\partial}{\partial t} \tag{11}$$

1.2.2 The Schrödinger Equation

Considering conservation of energy:

$$\hat{H} = \frac{\hat{p}^2}{2m} + V \tag{12}$$

We got The Schrödinger Equation:

Time-dependent:

$$i\hbar\frac{\partial\Psi}{\partial t} = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r},t)\right]\Psi. \tag{13}$$

Time-independent:

$$\Psi(\mathbf{r},t) = \psi(\mathbf{r})e^{-iEt/\hbar}, \qquad E\psi(\mathbf{r}) = \left[-\frac{\hbar^2}{2m}\nabla^2 + V(\mathbf{r})\right]\psi(\mathbf{r}). \tag{14}$$

1.2.3 Commutation Relations

$$[\hat{x},\hat{p}_x] = i\hbar, \qquad [\hat{L}_i,\hat{L}_j] = i\hbar\varepsilon_{ijk}\hat{L}_k. \tag{15}$$

1.2.4 Heisenberg Uncertainty Principle

$$\Delta x\,\Delta p_x \geq \frac{\hbar}{2}, \qquad \Delta E\,\Delta t \geq \frac{\hbar}{2}. \tag{16}$$

1.2.5 Representation Transformation

$$|\Psi\rangle = \sum_i a_i |u_i\rangle = \sum_j b_j |v_j\rangle , \tag{17}$$

$$b_j = \sum_i S_{ji} a_i, \qquad S_{ji} = \langle v_j | u_i \rangle. \tag{18}$$

Momentum and position representations:

$$\langle p|\Psi\rangle = \Psi(p), \qquad \langle x|\Psi\rangle = \Psi(x), \tag{19}$$

$$\Psi(p) = \int \langle p|x\rangle \Psi(x)\ \mathrm{d}x, \qquad \langle x|p\rangle = \frac{1}{\sqrt{2\pi\hbar}} e^{ipx/\hbar} = \psi_p(x). \tag{20}$$

1.3 Ehrenfest Theorem

Ehrenfest Theorem is the connection between quantum mechanics and classical mechanics.

For any operator $\hat{A}$ with no explicit time dependence:

$$\frac{d}{dt}\langle \hat{A} \rangle = \frac{1}{i\hbar} \langle [\hat{A}, \hat{H}] \rangle. \tag{21}$$

In particular:

$$\frac{d}{dt}\langle x \rangle = \frac{\langle p \rangle}{m}, \qquad \frac{d}{dt}\langle p \rangle = -\left\langle \frac{\partial V}{\partial x} \right\rangle = \langle F \rangle. \tag{22}$$

This corresponds to classical Poisson brackets: $\frac{\partial A}{\partial t} + [A, H] = 0$.

2 1D Motion

2.1 The 1D Infinite Square Well

Potential:

$$V(x) = \begin{cases} 0, & 0 < x < L, \\ \infty, & \text{otherwise.} \end{cases} \tag{23}$$

Energy eigenvalues:

$$E_n = \frac{n^2\pi^2\hbar^2}{2mL^2}, \qquad n = 1, 2, 3, \ldots \tag{24}$$

Normalised eigenfunctions:

$$\psi_n(x) = \sqrt{\frac{2}{L}} \sin\left(\frac{n\pi x}{L}\right), \qquad 0 < x < L. \tag{25}$$

2.2 The 1D Harmonic Oscillator

Potential:

$$V(x) = \frac{1}{2}kx^2, \qquad \omega = \sqrt{\frac{k}{m}}, \tag{26}$$

$$\hat{H} = -\frac{\hbar^2}{2m}\frac{d^2}{dx^2} + \frac{1}{2}m\omega^2x^2. \tag{27}$$

Define

$$\xi = \sqrt{\frac{m\omega}{\hbar}}\, x, \tag{28}$$

$$\hat{a} = \frac{1}{\sqrt{2}}\left(\xi + \frac{\partial}{\partial\xi}\right), \tag{29}$$

$$\hat{a}^\dagger = \frac{1}{\sqrt{2}}\left(\xi - \frac{\partial}{\partial\xi}\right), \tag{30}$$

with

$$[\hat{a}, \hat{a}^\dagger] = 1. \tag{31}$$

Then

$$E\Psi(x) = \hat{H}\Psi(x) = \frac{1}{2}\hbar\omega\left(\xi^2 - \frac{\partial^2}{\partial\xi^2}\right)\Psi(x) = \hbar\omega\left(\hat{a}^\dagger\hat{a} + \frac{1}{2}\right)\Psi(x). \tag{32}$$

Ground state:

$$\hat{a}\,|E_0\rangle = 0 \implies E_0 = \frac{1}{2}\hbar\omega, \tag{33}$$

and

$$\hat{H}\hat{a}^\dagger\,|E_n\rangle = (E_n + \hbar\omega)\hat{a}^\dagger\,|E_n\rangle\,, \qquad \hat{H}\hat{a}\,|E_n\rangle = (E_n - \hbar\omega)\hat{a}\,|E_n\rangle\,. \tag{34}$$

Thus

$$E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \qquad |E_n\rangle = \frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^n\,|E_0\rangle\,, \tag{35}$$

$$\hat{a}\,|n\rangle = \sqrt{n}\;|n-1\rangle\,, \qquad \hat{a}^\dagger\,|n\rangle = \sqrt{n+1}\;|n+1\rangle\,. \tag{36}$$

The ground state wavefunction satisfies

$$\hat{H}\psi_0(x) = E_0\psi_0(x) = \frac{1}{2}\hbar\omega\,\psi_0(x), \tag{37}$$

and

$$0 = \hat{a}\psi_0 = \frac{1}{\sqrt{2}}\left(\xi + \frac{\partial}{\partial\xi}\right)\psi_0, \tag{38}$$

with normalisation $\int_{-\infty}^{\infty}|\psi_0|^2dx = 1$, giving

$$\psi_0(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\xi^2/2}. \tag{39}$$

Excited states:

$$\begin{aligned}\psi_n(x) &= \frac{1}{\sqrt{n!}}(\hat{a}^\dagger)^n\psi_0 = \frac{1}{\sqrt{n!}}\left[\frac{1}{\sqrt{2}}\left(\xi - \frac{\partial}{\partial\xi}\right)\right]^n\left(\frac{m\omega}{\pi\hbar}\right)^{1/4} e^{-\xi^2/2} \\ &= \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\frac{1}{\sqrt{2^n n!}}\left(\xi - \frac{\partial}{\partial\xi}\right)^n e^{-\xi^2/2} = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\frac{1}{\sqrt{2^n n!}}H_n(\xi)e^{-\xi^2/2}.\end{aligned} \tag{40}$$

Here $H_n(\xi)$ are Hermite polynomials:

$$H_0(\xi) = 1, \quad H_1(\xi) = 2\xi, \quad H_2(\xi) = 4\xi^2 - 2, \quad \ldots \tag{41}$$

Final result:

$$\boxed{E_n = \left(n + \frac{1}{2}\right)\hbar\omega, \qquad \psi_n(x) = \left(\frac{m\omega}{\pi\hbar}\right)^{1/4}\frac{1}{\sqrt{2^n n!}}H_n(\xi)e^{-\xi^2/2}.} \tag{42}$$

3 3D Motion, Angular Momentum & Spin

3.1 Two-Body Problem

Centre-of-mass and relative coordinates:

$$\mathbf{R} = \frac{m_1\mathbf{r}_1 + m_2\mathbf{r}_2}{m_1 + m_2}, \tag{43}$$

$$\mathbf{r} = \mathbf{r}_2 - \mathbf{r}_1, \tag{44}$$

$$M = m_1 + m_2, \qquad \mu = \frac{m_1 m_2}{m_1 + m_2}, \tag{45}$$

$$\mathbf{r}_1 = \mathbf{R} - \frac{m_2}{m_1 + m_2}\mathbf{r}, \qquad \mathbf{r}_2 = \mathbf{R} + \frac{m_1}{m_1 + m_2}\mathbf{r}. \tag{46}$$

Momenta:

$$\mathbf{P} = M\dot{\mathbf{R}}, \qquad \mathbf{p} = \mu\dot{\mathbf{r}}, \qquad \mathbf{p}_1 = m_1\dot{\mathbf{r}}_1, \qquad \mathbf{p}_2 = m_2\dot{\mathbf{r}}_2. \tag{47}$$

Total energy:

$$E = \frac{p_1^2}{2m_1} + \frac{p_2^2}{2m_2} + V(r) = \frac{P^2}{2M} + \frac{p^2}{2\mu} + V(r). \tag{48}$$

Quantisation:

$$\hat{H}_C\psi_C(\mathbf{R}) = \frac{\hat{P}^2}{2M}\psi_C(\mathbf{R}) = E_C\psi_C(\mathbf{R}), \tag{49}$$

$$\hat{H}_r\psi_r(\mathbf{r}) = \frac{\hat{p}^2}{2\mu}\psi_r(\mathbf{r}) + V(r)\psi_r(\mathbf{r}) = E_r\psi_r(\mathbf{r}), \tag{50}$$

$$\Psi = \psi_C(\mathbf{R})\psi_r(\mathbf{r}), \tag{51}$$

$$\hat{P} = -i\hbar\nabla_R, \qquad \hat{p} = -i\hbar\nabla_r. \tag{52}$$

3.2 Angular Momentum and Spin

$$\hat{\mathbf{L}} = \mathbf{r} \times \hat{\mathbf{p}}, \tag{53}$$

so

$$\hat{L}_x = y\hat{p}_z - z\hat{p}_y, \quad \hat{L}_y = z\hat{p}_x - x\hat{p}_z, \quad \hat{L}_z = x\hat{p}_y - y\hat{p}_x. \tag{54}$$

In spherical coordinates ($x = r\sin\theta\cos\phi,\ y = r\sin\theta\sin\phi,\ z = r\cos\theta$):

$$\hat{L}_x = i\hbar\left(\sin\phi\frac{\partial}{\partial\theta} + \cot\theta\cos\phi\frac{\partial}{\partial\phi}\right), \tag{55}$$

$$\hat{L}_y = i\hbar\left(-\cos\phi\frac{\partial}{\partial\theta} + \cot\theta\sin\phi\frac{\partial}{\partial\phi}\right), \tag{56}$$

$$\hat{L}_z = -i\hbar\frac{\partial}{\partial\phi}, \tag{57}$$

$$\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]. \tag{58}$$

Define

$$\hat{L}^2 = \hat{L}_x^2 + \hat{L}_y^2 + \hat{L}_z^2, \qquad \hat{L}_\pm = \hat{L}_x \pm i\hat{L}_y, \tag{59}$$

and spin operators $\hat{\mathbf{S}} = \frac{\hbar}{2}\hat{\sigma}$, with Pauli matrices

$$\hat{\sigma}_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \quad \hat{\sigma}_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}, \quad \hat{\sigma}_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}, \tag{60}$$

satisfying

$$\hat{\sigma}_x^2 = \hat{\sigma}_y^2 = \hat{\sigma}_z^2 = \hat{I}, \qquad [\hat{\sigma}_i, \hat{\sigma}_j] = 2i\varepsilon_{ijk}\hat{\sigma}_k. \tag{61}$$

Commutation relations for $\hat{L}$:

$$[\hat{L}_x, \hat{L}_y] = i\hbar\hat{L}_z, \tag{62}$$
$$[\hat{L}_y, \hat{L}_z] = i\hbar\hat{L}_x, \tag{63}$$
$$[\hat{L}_z, \hat{L}_x] = i\hbar\hat{L}_y, \tag{64}$$
$$[\hat{L}^2, \hat{L}_z] = 0. \tag{65}$$

Eigenvalues:

$$\hat{L}^2 \left|lm\right\rangle = l(l+1)\hbar^2 \left|lm\right\rangle, \tag{66}$$
$$\hat{L}_z \left|lm\right\rangle = m\hbar \left|lm\right\rangle, \tag{67}$$
$$\hat{L}_\pm \left|lm\right\rangle = \hbar\sqrt{l(l+1) - m(m\pm 1)}\ \left|l, m\pm 1\right\rangle, \tag{68}$$

with

$$l = 0, 1, 2, \ldots, \qquad m = -l, -l+1, \ldots, l. \tag{69}$$

3.3 Stern–Gerlach Experiment

A beam of spin-1/2 particles in a non-uniform magnetic field splits into two discrete paths, demonstrating quantisation of spin.

3.4 Molecular Rotational Energy Levels

For a diatomic molecule with reduced mass μ, moment of inertia $I = \mu r^2$:

$$\hat{L}^2 = -\hbar^2\left[\frac{1}{\sin\theta}\frac{\partial}{\partial\theta}\left(\sin\theta\frac{\partial}{\partial\theta}\right) + \frac{1}{\sin^2\theta}\frac{\partial^2}{\partial\phi^2}\right]. \tag{70}$$

The full Hamiltonian:

$$\hat{H}_r = -\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{\hat{L}^2}{2\mu r^2} + V(r). \tag{71}$$

If the potential is harmonic $V(r) = \frac{1}{2}m\omega^2 r^2$, then for vibrational and rotational modes:

$$\begin{aligned}\hat{H}_r\psi_{nl} &= \left[-\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) + \frac{\hat{L}^2}{2\mu r^2} + V(r)\right]\psi_{nl} \\ &= \left[\left(n+\frac{1}{2}\right)\hbar\omega + \frac{l(l+1)\hbar^2}{2I}\right]\psi_{nl}.\end{aligned} \tag{72}$$

Thus

$$E_{n+1} - E_n = \hbar\omega = h\nu, \qquad E_{l+1} - E_l = \frac{\hbar^2}{I}. \tag{73}$$

3.5 Clebsch–Gordan Coefficients

$$|j_1, j_2; j, m\rangle = \sum_{m_1+m_2=m} C^{jm}_{j_1 m_1 j_2 m_2} \, |j_1, m_1; j_2, m_2\rangle \,, \tag{74}$$

$$C^{jm}_{j_1 m_1 j_2 m_2} = (-1)^{j_1 - j_2 + m} \sqrt{2j+1} \begin{pmatrix} j_1 & j_2 & j \\ m_1 & m_2 & -m \end{pmatrix} = \langle j_1, m_1; j_2, m_2 | j_1, j_2, j, m \rangle. \tag{75}$$

Allowed values:

$$\begin{cases} j = |j_1 - j_2|,\ |j_1 - j_2| + 1, \ldots, j_1 + j_2 - 1, j_1 + j_2, \\ m = -j, -j+1, \ldots, j-1, j, \\ m_1 = -j_1, \ldots, j_1, \quad m_2 = -j_2, \ldots, j_2. \end{cases} \tag{76}$$

The 3-j symbol:

$$\begin{pmatrix} j_1 & j_2 & j_3 \\ m_1 & m_2 & m_3 \end{pmatrix} = \delta_{m_1+m_2+m_3,0} \, \Delta(j_1, j_2, j_3) \sqrt{(j_1+m_1)!(j_1-m_1)!(j_2+m_2)!(j_2-m_2)!(j_3+m_3)!(j_3-m} \\ \times \sum_k \frac{(-1)^k}{k!(j_1+j_2-j_3-k)!(j_1-m_1-k)!(j_2+m_2-k)!(j_3-j_2+m_1+k)!(j_3-j_1-} \tag{77}$$

with

$$\Delta(j_1, j_2, j_3) = \sqrt{\frac{(j_1+j_2-j_3)!(j_1-j_2+j_3)!(-j_1+j_2+j_3)!}{(j_1+j_2+j_3+1)!}}. \tag{78}$$

In practice, coefficients are often looked up in tables.

4 Hydrogen Atom & Stationary Perturbation Theory

4.1 Spherically Symmetric Potential and the Hydrogen Atom

4.1.1 General radial equation

For $V = V(r)$:

$$\hat{H}_r = -\frac{\hbar^2}{2\mu}\left[\frac{1}{r^2}\frac{\partial}{\partial r}\left(r^2\frac{\partial}{\partial r}\right) - \frac{\hat{L}^2}{\hbar^2 r^2}\right] + V(r), \tag{79}$$

$$\Psi_E(r,\theta,\phi) = R(r)Y_l(\theta,\phi). \tag{80}$$

Then

$$\hat{L}^2 Y_l = l(l+1)\hbar^2 Y_l, \tag{81}$$

$$\left[-\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d}{dr}\right) + V(r) + \frac{l(l+1)\hbar^2}{2\mu r^2}\right] R(r) = ER(r). \tag{82}$$

4.1.2 Angular solutions

$$Y_l(\theta,\phi) = \Theta(\theta)\Phi(\phi), \tag{83}$$

$$\hat{L}_z^2\Phi = -\hbar^2\frac{d^2\Phi}{d\phi^2} = m^2\hbar^2\Phi \implies \Phi(\phi) \propto e^{im\phi}, \quad m = -l,\ldots,l, \tag{84}$$

$$\left[-\frac{\hbar^2}{\sin\theta}\frac{d}{d\theta}\left(\sin\theta\frac{d}{d\theta}\right) + \frac{m^2\hbar^2}{\sin^2\theta}\right]\Theta = l(l+1)\hbar^2\Theta, \tag{85}$$

so $\Theta(\theta) = P_l^m(\cos\theta)$ (associated Legendre polynomials).

Normalised spherical harmonics:

$$Y_l^m(\theta,\phi) = (-1)^{(m+|m|)/2}\sqrt{\frac{(2l+1)(l-|m|)!}{4\pi(l+|m|)!}}\,P_l^m(\cos\theta)\,e^{im\phi}, \tag{86}$$

with

$$Y_l^{-m}(\theta,\phi) = (-1)^m\left[Y_l^m(\theta,\phi)\right]^*. \tag{87}$$

4.1.3 Hydrogen atom –zeroth order

Coulomb potential:

$$V(r) = -\frac{e^2}{4\pi\varepsilon_0 r} = -\frac{\alpha\hbar c}{r}, \qquad \alpha = \frac{e^2}{4\pi\varepsilon_0\hbar c} \approx \frac{1}{137}. \tag{88}$$

The radial equation becomes

$$0 = \left[-\frac{\hbar^2}{2\mu}\frac{1}{r^2}\frac{d}{dr}\left(r^2\frac{d}{dr}\right) - \frac{\alpha\hbar c}{r} + \frac{l(l+1)\hbar^2}{2\mu r^2} - E\right]R, \tag{89}$$

or

$$\frac{d^2R}{dr^2} + \frac{2}{r}\frac{dR}{dr} + \left[\frac{2\mu E}{\hbar^2} + \frac{2\mu\alpha}{\hbar r} - \frac{l(l+1)}{r^2}\right]R = 0. \tag{90}$$

Asymptotic behaviour:

- $r \to \infty$: $R(r) \propto e^{-\sqrt{2\mu|E|}\, r/\hbar}$ (bound states, $E<0$).
- $r \to 0$: $R(r) \propto r^{l}$.

Define

$$A = \sqrt{\frac{2\mu|E|}{\hbar^2}}, \qquad B = \frac{2\mu c\alpha}{\hbar}, \qquad u(r) = R(r)r. \tag{91}$$

Then

$$u(r) = e^{-Ar} r^{l+1} F(r), \qquad \frac{u''}{u} + \left[-A^2 + \frac{B}{r} - \frac{l(l+1)}{r^2}\right] = 0. \tag{92}$$

With $F(r) = \sum_{i=0}^{\infty} a_i r^i$, one obtains the recurrence

$$\frac{a_{i+1}}{a_i} = \frac{2A\left[i + l + 1 - \frac{B}{2A}\right]}{(i+1)(i+2l+2)}. \tag{93}$$

Quantisation requires $n = \frac{B}{2A} = \alpha c\sqrt{\frac{\mu}{2|E|}}$ with $l + 1 - n \le 0$. Hence

$$E_n = -\frac{\mu c^2 \alpha^2}{2n^2}, \qquad A = \frac{\mu c\alpha}{n\hbar}, \qquad n = 1, 2, \ldots \tag{94}$$

Bohr radius:

$$a_0 = \frac{\hbar}{\mu c\alpha} = \frac{4\pi\varepsilon_0\hbar^2}{\mu e^2} \approx 0.529\,\text{Å}. \tag{95}$$

Radial wavefunctions:

$$R_{nl}(r) \propto e^{-r/na_0} \left(\frac{2r}{na_0}\right)^{l} L_{n+l}^{2l+1}\left(\frac{2r}{na_0}\right), \tag{96}$$

where L_{n+l}^{2l+1} are associated Laguerre polynomials. Normalised:

$$R_{nl}(r) = -\sqrt{\left(\frac{2}{na_0}\right)^3 \frac{(n-l-1)!}{2n[(n+l)!]^3}}\, e^{-r/na_0} \left(\frac{2r}{na_0}\right)^{l} L_{n+l}^{2l+1}\left(\frac{2r}{na_0}\right). \tag{97}$$

Full hydrogen wavefunctions:

$$\Psi_{nlm} = R_{nl}(r) Y_l^m(\theta, \phi). \tag{98}$$

Explicit lowest states:

$$\Psi_{100} = \frac{1}{\sqrt{\pi}} \left(\frac{1}{a_0}\right)^{3/2} e^{-r/a_0}, \tag{99}$$

$$\Psi_{200} = \frac{1}{4\sqrt{2\pi}} \left(\frac{1}{a_0}\right)^{3/2} \left(2 - \frac{r}{a_0}\right) e^{-r/2a_0}, \tag{100}$$

$$\Psi_{210} = \frac{1}{4\sqrt{2\pi}} \left(\frac{1}{a_0}\right)^{3/2} \frac{r}{a_0} e^{-r/2a_0} \cos\theta, \tag{101}$$

$$\Psi_{21,\pm1} = \mp\frac{1}{8\sqrt{\pi}} \left(\frac{1}{a_0}\right)^{3/2} \frac{r}{a_0} e^{-r/2a_0} \sin\theta\, e^{\pm i\phi}. \tag{102}$$

Most probable radius (for circular orbits $l = n-1$) is $r_{\max} \approx n^2 a_0$.

Energy levels:

$$E_n^{(0)} = -\frac{m_e c^2 \alpha^2}{2n^2} = -\frac{m_e e^4}{8\varepsilon_0^2 \hbar^2}\frac{1}{n^2} = -\frac{13.6\text{ eV}}{n^2}. \tag{103}$$

Degeneracy including spin:

$$\text{degeneracy} = 2\sum_{l=0}^{n-1}(2l+1) = 2n^2. \tag{104}$$

Spin-orbit coupling (to be added in fine structure):

$$\hat{H}_{so} = \xi(r)\hat{\mathbf{L}}\cdot\hat{\mathbf{S}}, \qquad \xi(r) = \frac{1}{2m^2c^2}\frac{e^2}{4\pi\varepsilon_0}\frac{1}{r^3}. \tag{105}$$

4.2 Time-Independent Perturbation Theory

4.2.1 Non-degenerate case

Given $\hat{H} = \hat{H}_0 + \lambda\hat{H}'$, with known $E_n^{(0)}, |n^{(0)}\rangle$:

$$E_n = E_n^{(0)} + \lambda E_n^{(1)} + \lambda^2 E_n^{(2)} + \cdots, \tag{106}$$

$$|n\rangle = |n^{(0)}\rangle + \lambda|n^{(1)}\rangle + \lambda^2|n^{(2)}\rangle + \cdots, \tag{107}$$

and $\langle n^{(l)}|n^{(k)}\rangle = \delta_{lk}$.

Order-by-order equations:

$$0\text{th}: \quad (\hat{H}_0 - E_n^{(0)})|n^{(0)}\rangle = 0, \tag{108}$$

$$1\text{st}: \quad (\hat{H}_0 - E_n^{(0)})|n^{(1)}\rangle + (\hat{H}' - E_n^{(1)})|n^{(0)}\rangle = 0, \tag{109}$$

$$2\text{nd}: \quad (\hat{H}_0 - E_n^{(0)})|n^{(2)}\rangle + (\hat{H}' - E_n^{(1)})|n^{(1)}\rangle + E_n^{(2)}|n^{(0)}\rangle = 0, \tag{110}$$

and generally for the l-th order:

$$(\hat{H}_0 - E_n^{(0)})|n^{(l)}\rangle + (\hat{H}' - E_n^{(1)})|n^{(l-1)}\rangle + E_n^{(2)}|n^{(l-2)}\rangle + \cdots + E_n^{(l)}|n^{(0)}\rangle = 0. \tag{111}$$

Define $H'_{nm} = \langle n^{(0)}|\hat{H}'|m^{(0)}\rangle$. Then

$$E_n^{(1)} = H'_{nn}, \qquad E_n^{(2)} = \sum_{m\neq n}\frac{|H'_{nm}|^2}{E_n^{(0)} - E_m^{(0)}}. \tag{112}$$

Also

$$|n^{(1)}\rangle = \sum_{m\neq n}\frac{H'_{mn}}{E_n^{(0)} - E_m^{(0)}}|m^{(0)}\rangle. \tag{113}$$

4.2.2 Degenerate case

Let the unperturbed subspace have dimension g_n with states $|n^{(0)}, \nu\rangle$, $\nu = 1, \ldots, g_n$. To first order in λ:

$$\hat{H}(\lambda) = \hat{H}_0 + \lambda\hat{H}', \tag{114}$$

$$E_k = E_k^{(0)} + \lambda E_k^{(1)}, \tag{115}$$

$$|k\rangle = |k^{(0)}\rangle + \lambda|k^{(1)}\rangle. \tag{116}$$

Define

$$H'_{n\nu,m\mu} = \langle n^{(0)}, \nu | \hat{H}' | m^{(0)}, \mu \rangle. \tag{117}$$

The first-order correction is found by diagonalising $\hat{H}'$ within the degenerate subspace. The secular equation:

$$\det\left(H'_{k\nu,k\mu} - E_k^{(1)}\delta_{\nu\mu}\right) = 0, \tag{118}$$

which yields g_k eigenvalues $E_{k,\alpha}^{(1)}$.

4.3 Fine Structure of the Hydrogen Atom

Corrections of order $\alpha^4 m_e c^2$.

4.3.1 Relativistic correction

$$E = \sqrt{m_e^2c^4 + p^2c^2} \approx m_e c^2 + \frac{p^2}{2m_e} - \frac{1}{8}\frac{p^4}{m_e^3c^2}. \tag{119}$$

Thus

$$\hat{H}_0 = \frac{\hat{p}^2}{2m_e} - \frac{\alpha\hbar c}{r}, \qquad \hat{H}'_{rel} = -\frac{1}{8}\frac{\hat{p}^4}{m_e^3c^2} = -\frac{1}{2m_ec^2}\left(\hat{H}_0 + \frac{\alpha\hbar c}{r}\right)^2. \tag{120}$$

Useful expectation values:

$$\left\langle nlm \left| \frac{\alpha\hbar c}{r} \right| nlm \right\rangle = -2E_n^{(0)} \quad \text{(virial theorem)}, \tag{121}$$

$$\left\langle nlm \left| \left(\frac{\alpha\hbar c}{r}\right)^2 \right| nlm \right\rangle = \frac{4n}{l+\frac{1}{2}}\left[E_n^{(0)}\right]^2. \tag{122}$$

Then

$$E_{n,rel}^{(1)} = -\frac{1}{2m_ec^2}\left[E_n^{(0)}\right]^2\left(-3 + \frac{4n}{l+\frac{1}{2}}\right) = -\frac{1}{2}m_ec^2\alpha^4\left[-\frac{3}{4n^4} + \frac{1}{n^3(l+\frac{1}{2})}\right]. \tag{123}$$

4.3.2 Spin-orbit interaction(S-L)

e spin - e orbit, perturbation Hamiltonian scaling $\hat{H}'_{\mathrm{SO}} \sim \alpha^4 m_e c^2$

Classical derivation of orbital magnetic moment Definition of magnetic moment for current loop:

$$\mu_L = IA = I\pi r^2 \tag{124}$$

Current generated by orbiting electron:

$$I = \frac{e}{T} = \frac{ev}{2\pi r} = \frac{e}{2\pi m_e r^2}L \tag{125}$$

Combine the two equations to eliminate current, obtain the magnitude relation of orbital magnetic moment:

$$\mu_L = \frac{e}{2m_e}L \tag{126}$$

Electron spin magnetic moment, non-relativistic approximation for electron spin g-factor $g_e = 2$:

$$\mu_S = g_e\frac{e}{2m_e}\cdot\boldsymbol{S}, \quad g_e = 2 \tag{127}$$

Biot–Savart's Law Remark: $\boldsymbol{B}$ due to proton

$$B = \oint \frac{\mu_0}{4\pi} \frac{I d\boldsymbol{\ell} \times \hat{r}}{r^2} \tag{128}$$

$$= \frac{I}{4\pi\varepsilon_0 c^2 r^2} \oint d\ell \tag{129}$$

$$= \frac{I \cdot 2\pi r}{4\pi\varepsilon_0 c^2 r^2} \tag{130}$$

$$= \frac{I}{2\varepsilon_0 c^2 r} \tag{131}$$

Vector Operator System of coupled equations for spin magnetic moment operator and magnetic field operator:

$$\begin{cases} \hat{\boldsymbol{\mu}}_S = -g_e \dfrac{e}{2m_e} \hat{\boldsymbol{S}} \\ \hat{\boldsymbol{B}} = \dfrac{e}{4\pi\varepsilon_0 m_e c^2 r^3} \hat{\boldsymbol{L}} \end{cases} \tag{132}$$

Spin-orbit perturbation Hamiltonian $\hat{H}'_{\mathrm{SO}} = -\hat{\boldsymbol{\mu}}_S \cdot \hat{\boldsymbol{B}}$. Substitute the operators and correct the typo $m_e \to m_e^2$ in denominator:

$$\hat{H}'_{\mathrm{SO}} = -\hat{\boldsymbol{\mu}}_S \cdot \hat{\boldsymbol{B}} \tag{133}$$

$$= g_e \cdot \frac{e^2}{8\pi\varepsilon_0 m_e^2 c^2 r^3} \left(\hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{S}} \right) \tag{134}$$

Substitute the approximation $g_e = 2$.

Total angular momentum identity:

$$\hat{\boldsymbol{J}} = \hat{\boldsymbol{L}} + \hat{\boldsymbol{S}} \Rightarrow \hat{\boldsymbol{L}} \cdot \hat{\boldsymbol{S}} = \frac{1}{2}\left(\hat{J}^2 - \hat{L}^2 - \hat{S}^2\right)$$

Substitute the dot product of angular momentum into the Hamiltonian:

$$\hat{H}'_{\mathrm{SO}} = \frac{e^2}{8\pi\varepsilon_0 m_e^2 c^2 r^3} \left(\hat{J}^2 - \hat{L}^2 - \hat{S}^2 \right) \tag{135}$$

Fine-structure constant and energy scaling estimation Definition of fine-structure constant, equivalent form of Bohr radius, scaling of squared angular momentum operators:

$$\because \alpha = \frac{e^2}{4\pi\varepsilon_0 \hbar c}, \quad \langle r \rangle \sim a_0 = \frac{\hbar}{\alpha \cdot m_e \cdot c}, \quad \hat{J}^2, \hat{L}^2, \hat{S}^2 \sim \hbar^2$$

Scaling of expectation value of spin-orbit interaction energy:

$$\langle \hat{H}'_{\mathrm{SO}} \rangle \sim \frac{1}{2}\alpha^4 \cdot m_e c^2 \sim \alpha^2 \left| E_n^{(0)} \right| \tag{136}$$

the first-order energy shift is

$$E^{(1)}_{nlj,so} = -\frac{Z^2\alpha^2}{2nl(l+1)(l+2)} E_n^{(0)} \begin{cases} l, & j = l + \frac{1}{2}, \\ -(l+1), & j = l - \frac{1}{2}. \end{cases} \tag{137}$$

4.3.3 Lamb shift

Dirac theory predicts degeneracy between $2S_{1/2}$ and $2P_{1/2}$. The Lamb shift (order $\alpha^5 m_e c^2$) lifts this:

- **Electron self-energy** ($\approx$97%):

$$\Delta E_{SE}(2S_{1/2}) = \frac{\alpha^5 m_e c^2}{6\pi n^3}\left(\ln\frac{1}{\alpha^2} - \frac{19}{30}\right)\frac{4(2\alpha)^4 m_e c^2}{32}, \tag{138}$$

$$\Delta E_{SE}(2P_{1/2}) \approx 0. \tag{139}$$

- **Vacuum polarisation** ($\approx$3%):

$$\Delta E_{VP}(2S_{1/2}) = \left(\frac{\alpha^5 m_e c^2}{15\pi n^3}\right)\frac{\alpha}{\pi}\frac{(2\alpha)^4}{n^3}F_{VP}\, m_e c^2, \tag{140}$$

$$\Delta E_{VP}(2P_{1/2}) \approx 0. \tag{141}$$

For $n = 2,\ Z = 1,\ \alpha = 1/137.036,\ m_e = 0.511$ MeV:

$$\Delta E_{\text{Lamb}} = E(2S_{1/2}) - E(2P_{1/2}) \approx 4.37 \times 10^{-6}\ \text{eV}, \tag{142}$$

$$\nu = \frac{\Delta E_{\text{Lamb}}}{h} \approx 1057.8\ \text{MHz}. \tag{143}$$

4.4 Hyperfine Structure of the Hydrogen Atom (I-S)

Interaction between electron spin and proton spin; order $\alpha^4 \frac{m_e}{m_p} m_e c^2$.

Electron magnetic moment:

$$\hat{\mu}_s = -g_e \frac{\mu_B}{\hbar}\hat{\mathbf{S}}, \qquad \mu_B = \frac{e\hbar}{2m_e}. \tag{144}$$

Proton magnetic moment:

$$\hat{\mu}_p = g_p \frac{\mu_N}{\hbar}\hat{\mathbf{I}}, \qquad \mu_N = \frac{e\hbar}{2m_p}. \tag{145}$$

Magnetic field due to proton spin at the origin:

$$\hat{\mathbf{B}}_p(r) \approx \frac{2}{3}\mu_0\, \delta^3(r)\, \hat{\mu}_p. \tag{146}$$

Hyperfine Hamiltonian:

$$\hat{H}'_{hfs} = -\hat{\mathbf{B}}_p \cdot \hat{\mu}_s = \frac{2}{3}\mu_0 \frac{g_e g_p \mu_N \mu_B}{\hbar^2}(\hat{\mathbf{I}}\cdot\hat{\mathbf{S}})\,\delta^3(r). \tag{147}$$

Total spin $\hat{\mathbf{F}} = \hat{\mathbf{S}} + \hat{\mathbf{I}}$, with

$$\hat{F}^2|FM_F\rangle = F(F+1)\hbar^2|FM_F\rangle, \qquad \hat{F}_z|FM_F\rangle = M_F\hbar|FM_F\rangle. \tag{148}$$

Then

$$\hat{\mathbf{I}}\cdot\hat{\mathbf{S}} = \frac{1}{2}(\hat{F}^2 - \hat{I}^2 - \hat{S}^2). \tag{149}$$

For the $1s$ state, $|\psi_{1s}(0)|^2 = \frac{1}{\pi a_0^3}$. Hence

$$E_{hfs,1s}(F) = \frac{1}{3}\mu_0 g_e g_p \mu_N \mu_B \left[F(F+1) - \frac{3}{2}\right] |\psi_{1s}(0)|^2, \tag{150}$$

since $S = I = 1/2$, so $F = 0, 1$.

The splitting between $F = 1$ and $F = 0$:

$$\Delta E_{hfs,1s} = E(F=1) - E(F=0) = \frac{2}{3\pi a_0^3}\mu_0 g_e g_p \mu_N \mu_B = \frac{hc}{\lambda_{hfs}}. \tag{151}$$

Using $a_0 = 5.3 \times 10^{-11}$ m and constants $\mu_N = 3.15 \times 10^{-8}$ eV/T, $\mu_B = 5.8 \times 10^{-5}$ eV/T, $g_e \approx 2$, $g_p \approx 5.6$, $\mu_0 = 4\pi \times 10^{-7}$ N/A^2, one obtains

$$\Delta E_{hfs,1s} \approx 6 \times 10^{-6} \text{ eV}, \qquad \lambda_{hfs} \approx 21 \text{ cm}. \tag{152}$$

5 Atoms in External Magnetic Fields

5.1 Electron Spin Resonance (B-S)

For a free electron in $\mathbf{B} = B\hat{z}$:

$$\hat{H}'_{ESR} = g_e \frac{\mu_B}{\hbar} \hat{S}_z B, \tag{153}$$

so

$$E_{ESR} = g_e \mu_B m_s B, \qquad m_s = \pm\frac{1}{2}, \tag{154}$$

and the transition frequency:

$$h\nu = g_e \mu_B B \implies \nu = \frac{g_e \mu_B B}{h}. \tag{155}$$

5.2 Nuclear Magnetic Resonance (B-I)

For a proton:

$$E_{NMR} = -g_p \mu_N m_I B, \qquad m_I = \pm\frac{1}{2}, \tag{156}$$

$$h\nu = g_p \mu_N B \implies \nu = \frac{g_p \mu_N B}{h}. \tag{157}$$

5.3 Normal Zeeman effect (B-L)

$$\hat{H}'_{NZ} = \frac{\mu_B}{\hbar} \hat{L}_z B, \qquad E_{NZ} = \mu_B m_l B. \tag{158}$$

For a transition $2P \to 1S$:

$$h\nu = \frac{3}{8} m_e c^2 \alpha^2 + \mu_B m_l B, \tag{159}$$

so with $\nu_0 = \frac{3 m_e c^2 \alpha^2}{8h}$:

$$\nu = \nu_0 + \frac{\mu_B B}{h} \ (m_l = 1), \quad \nu = \nu_0 \ (m_l = 0), \quad \nu = \nu_0 - \frac{\mu_B B}{h} \ (m_l = -1). \tag{160}$$

5.4 Anomalous Zeeman effect (weak B - L+S coupling)

$$\hat{\mu} = -\frac{\mu_B}{\hbar}(\hat{\mathbf{L}} + 2\hat{\mathbf{S}}) \implies \hat{\mu} = -g\frac{\mu_B}{\hbar}\hat{\mathbf{J}}, \tag{161}$$

with

$$g = 1 + \frac{j(j+1) + s(s+1) - l(l+1)}{2j(j+1)}. \tag{162}$$

Then

$$E_{AZ} = g \mu_B m_j B. \tag{163}$$

For example:

- ${}^2S_{1/2}$: $g = 2$, $m_j = \pm\frac{1}{2}$.
- ${}^2P_{1/2}$: $g = \frac{2}{3}$, $m_j = \pm\frac{1}{2}$.
- ${}^2P_{3/2}$: $g = \frac{4}{3}$, $m_j = -\frac{3}{2}, -\frac{1}{2}, \frac{1}{2}, \frac{3}{2}$.

For the $2P_{1/2} \to 1S_{1/2}$ transition, four lines appear at

$$\nu = \nu_0 \pm \frac{2}{3}\frac{\mu_B B}{h}, \qquad \nu = \nu_0 \pm \frac{4}{3}\frac{\mu_B B}{h}. \tag{164}$$

5.5 Paschen–Back effect (strong B - L+S decoupled)

$$\hat{H}'_{PB} = \frac{\mu_B}{\hbar}(\hat{L}_z + 2\hat{S}_z)B, \tag{165}$$

so

$$E_{PB} = \mu_B B(m_l + 2m_s). \tag{166}$$

For a $2P \to 1S$ transition, selection rules $\Delta m_l = 0, \pm 1$, $\Delta m_s = 0$ give three lines:

$$\nu = \nu_0 - \frac{\mu_B B}{h}, \quad \nu_0, \quad \nu_0 + \frac{\mu_B B}{h}. \tag{167}$$

6 Many-Body Systems & Identical Particles

6.1 Identical Particles

- Bosons (integer spin): symmetric wavefunction.
- Fermions (half-integer spin): anti-symmetric wavefunction.

Spin states for two spin-1/2 particles:

- Symmetric triplet:

$$|11\rangle = |{+}{+}\rangle, \tag{168}$$

$$|10\rangle = \frac{1}{\sqrt{2}}(|{+}{-}\rangle + |{-}{+}\rangle), \tag{169}$$

$$|1\ -1\rangle = |{-}{-}\rangle. \tag{170}$$

- Anti-symmetric singlet:

$$|00\rangle = \frac{1}{\sqrt{2}}(|{+}{-}\rangle - |{-}{+}\rangle). \tag{171}$$

For fermions, the total wavefunction must be anti-symmetric:

$$|\Psi^A\rangle = |\Psi^A_{spatial}\rangle|\Psi^S_{spin}\rangle = |\Psi^S_{spatial}\rangle|\Psi^A_{spin}\rangle. \tag{172}$$

Interacting identical particles:

- Direct integral:

$$J_{nm} = \int \phi_n^*(\mathbf{x}_1)\phi_m^*(\mathbf{x}_2)V_{int}(|\mathbf{x}_2 - \mathbf{x}_1|)\phi_n(\mathbf{x}_1)\phi_m(\mathbf{x}_2)\, d\mathbf{x}_1 d\mathbf{x}_2. \tag{173}$$

- Exchange integral:

$$K_{nm} = \int \phi_n^*(\mathbf{x}_1)\phi_m^*(\mathbf{x}_2)V_{int}(|\mathbf{x}_2 - \mathbf{x}_1|)\phi_m(\mathbf{x}_1)\phi_n(\mathbf{x}_2)\, d\mathbf{x}_1 d\mathbf{x}_2. \tag{174}$$

- Bosons (spin 0): $\Delta E_{12} = J_{12} + K_{12}$.
- Fermions: symmetric spatial $\rightarrow J_{12} + K_{12}$; anti-symmetric spatial $\rightarrow J_{12} - K_{12}$.

6.2 Helium Atom

Hamiltonian:

$$\hat{H} = -\frac{\hbar^2}{2m_e}\nabla_1^2 - \frac{\hbar^2}{2m_e}\nabla_2^2 - \frac{2e^2}{4\pi\varepsilon_0 r_1} - \frac{2e^2}{4\pi\varepsilon_0 r_2} + \frac{e^2}{4\pi\varepsilon_0 r_{12}}. \tag{175}$$

Ground state: $|\Psi^S_{spatial}\rangle = |1s, 1s\rangle$, so total wavefunction is

$$|\Psi^A_{He}\rangle = |\Psi^S_{spatial}\rangle|\Psi^A_{spin}\rangle. \tag{176}$$

Excited states: spatial part must be anti-symmetric:

$$|\Psi^A_{spatial}\rangle = \frac{1}{\sqrt{2}}(|1s,2s\rangle - |2s,1s\rangle), \quad \frac{1}{\sqrt{2}}(|1s,2p\rangle - |2p,1s\rangle). \tag{177}$$

For a configuration like $1s\,3d$: $L = 2$, $S = 0$ or 1, so

$$J = \begin{cases} 2 & (S=0), \\ 1,2,3 & (S=1). \end{cases} \tag{178}$$

Allowed transitions require $\Delta S = 0$.

Pauli exclusion principle: no two fermions can occupy the same quantum state.

6.3 Bose–Einstein Condensation (BEC)

Any number of bosons can occupy the same state. For hydrogen atoms, total spin $F = I + S$ with $I = S = 1/2$ gives $F = 0, 1$, so H atoms are bosons.

For a gas at temperature T:

$$\frac{1}{2}m_H v^2 = \frac{3}{2}kT \implies p = m_H v = \sqrt{3m_H kT}, \tag{179}$$

$$\lambda = \frac{h}{p} = \frac{h}{\sqrt{3m_H kT}}. \tag{180}$$

When the de Broglie wavelength becomes comparable to inter-atom spacing, condensation occurs at $T_c \sim 10^{-6}$ K.

7 TIME-DEPENDENT PERTURBATION & TRANSITIONS

The logic of this section is as follows. We first write the time-dependent perturbation problem in the interaction picture and obtain the transition amplitude from the Dyson series. We then discuss constant perturbations, Fermi's golden rule, second-order matrix elements, harmonic perturbations, Rabi oscillations, and finally apply electric-dipole transitions to the hydrogen atom and state the selection rules.

7.1 Interaction Picture

For a total Hamiltonian $H_0 + V(t)$, define the interaction-picture state and perturbation operator by

$$|\Psi(t)\rangle_I = e^{iH_0 t}|\Psi(t)\rangle, \qquad V_I(t) = e^{iH_0 t}V(t)e^{-iH_0 t}. \tag{181}$$

The interaction-picture state then satisfies

$$i\frac{\partial}{\partial t}|\Psi(t)\rangle_I = V_I(t)|\Psi(t)\rangle_I, \qquad |\Psi(t)\rangle_I = U_I(t,t_0)|\Psi(t_0)\rangle_I = \sum_n c_n(t)|n\rangle. \tag{182}$$

7.2 Dyson Series

The interaction-picture time-evolution operator satisfies

$$\begin{cases} i\dfrac{\partial U_I(t,t_0)}{\partial t} = V_I(t)U_I(t,t_0), \\ U_I(t_0,t_0) = 1. \end{cases}$$

Iterating this equation gives the Dyson series

$$\begin{aligned} U_I(t,t_0) &= 1 - i\int_{t_0}^{t} dt_1\, V_I(t_1)U_I(t_1,t_0) \\ &= 1 - i\int_{t_0}^{t} dt_1\, V_I(t_1)\left[1 - i\int_{t_0}^{t_1} dt_2\, V_I(t_2)U_I(t_2,t_0)\right] \\ &= \cdots \\ &= \sum_{n=0}^{\infty}(-i)^n \int_{t_0}^{t} dt_1 \int_{t_0}^{t_1} dt_2 \cdots \int_{t_0}^{t_{n-1}} dt_n\, V_I(t_1)V_I(t_2)\cdots V_I(t_n). \end{aligned} \tag{183}$$

Equivalently, it can be written as the time-ordered exponential

$$U_I(t,t_0) = \mathcal{T}\exp\left[-i\int_{t_0}^{t} dt'\, V_I(t')\right].$$

7.3 Transition Probability

Take the initial state to be $|i\rangle$. Then

$$|\Psi(t_0)\rangle_I = |i\rangle, \qquad |\Psi(t)\rangle_I = U_I(t,t_0)|i\rangle = \sum_m c_m(t)|m\rangle.$$

The transition amplitude is therefore

$$c_n(t) = \langle n|U_I(t,t_0)|i\rangle. \tag{184}$$

Define the Bohr frequency and the perturbation matrix element by

$$\omega_{nm} = E_n - E_m, \qquad \langle n|V_I(t)|m\rangle = e^{i\omega_{nm}t}V_{nm}, \qquad V_{nm} = \langle n|V|m\rangle.$$

To second order, the time-evolution operator is approximated by

$$\begin{aligned} U_I(t,t_0) \approx 1 &- i\int_{t_0}^{t} dt_1\, V_I(t_1) \\ &+ (-i)^2 \int_{t_0}^{t} dt_1 \int_{t_0}^{t_1} dt_2\, V_I(t_1)V_I(t_2). \end{aligned}$$

Thus

$$\begin{aligned} c_n(t) &= \langle n|U_I(t,t_0)|i\rangle \\ &\approx \delta_{ni} - i\int_{t_0}^{t} dt_1\, e^{i\omega_{ni}t_1}V_{ni}(t_1) \\ &\quad + (-i)^2 \int_{t_0}^{t} dt_1 \int_{t_0}^{t_1} dt_2 \sum_m e^{i\omega_{nm}t_1}e^{i\omega_{mi}t_2}V_{nm}(t_1)V_{mi}(t_2). \end{aligned}$$

The transition probability is

$$P(i \to n) = |c_n(t)|^2.$$

7.4 Constant Perturbation

Consider

$$V(t) = \begin{cases} 0, & t < 0, \\ V, & t \geq 0, \end{cases} \qquad t_0 = 0.$$

The zeroth- and first-order amplitudes are

$$\begin{cases} c_n^{(0)} = \delta_{ni}, \\ c_n^{(1)} = -iV_{ni}\displaystyle\int_0^t dt_1\, e^{i\omega_{ni}t_1} = \frac{V_{ni}}{\omega_{ni}}\left(1 - e^{i\omega_{ni}t}\right). \end{cases} \tag{185}$$

Its squared modulus is

$$\begin{aligned} \left|c_n^{(1)}\right|^2 &= \frac{|V_{ni}|^2}{|\omega_{ni}|^2} \cdot 4\sin^2\frac{\omega_{ni}t}{2} \\ &= \frac{4|V_{ni}|^2}{|E_n - E_i|^2}\sin^2\left[\frac{(E_n - E_i)t}{2}\right] \\ &= |V_{ni}|^2\, t^2\, \mathrm{sinc}^2\left[\frac{(E_n - E_i)t}{2}\right], \end{aligned} \tag{186}$$

where $\mathrm{sinc}\, x = \sin x/x$.

7.5 Fermi's Golden Rule

For continuum final states, take the long-time limit. Using

$$\int_{-\infty}^{\infty} \mathrm{sinc}^2 y \, dy = \int_{-\infty}^{\infty} \left(\frac{\sin y}{y}\right)^2 dy = \pi,$$

and

$$\lim_{\alpha\to\infty} \frac{\alpha}{\pi} \mathrm{sinc}^2(\alpha x) = \delta(x),$$

one obtains

$$\lim_{t\to\infty} \frac{t}{2\pi} \mathrm{sinc}^2 \left[\frac{(E_n - E_i)t}{2}\right] = \delta(E_n - E_i).$$

Therefore

$$\lim_{t\to\infty} \left|c_n^{(1)}(t)\right|^2 = 2\pi \, \overline{|V_{ni}|^2} \, t \, \delta(E_n - E_i).$$

Let $\rho(E)$ be the density of final states, i.e. the number of states in the energy interval $(E, E + dE)$. In the long-time limit,

$$\begin{aligned} P(i \to n) &= \sum_{\substack{n \\ E_n \approx E_i}} \left|c_n^{(1)}\right|^2_{t\to\infty} = \int dE_n \, \rho(E_n) \left|c_n^{(1)}(t)\right|^2_{t\to\infty} \\ &= \int dE_n \, \rho(E_n) \, 2\pi \overline{|V_{ni}|^2} \, t \, \delta(E_n - E_i) \\ &= 2\pi \overline{|V_{ni}|^2} \rho(E_i) \, t. \end{aligned}$$

Writing the transition matrix element as M_{fi}, the transition rate per unit time is

$$\Gamma_{fi} = \frac{d}{dt} P(i \to f) = 2\pi |M_{fi}|^2 \rho(E_f), \qquad E_f \approx E_i. \tag{187}$$

Here the first-order transition matrix element is

$$M_{fi} = \langle f|V|i\rangle = V_{fi}.$$

The second-order correction gives

$$M_{fi} = V_{fi} + \sum_m \frac{V_{fm} V_{mi}}{E_i - E_m}, \qquad \Gamma_{fi} = 2\pi |M_{fi}|^2 \rho(E_f), \quad E_f \approx E_i. \tag{188}$$

7.6 Harmonic Perturbation

Take

$$V(t) = V e^{i\omega t} + V^\dagger e^{-i\omega t}, \qquad t_0 = 0.$$

The first-order amplitude is

$$\begin{aligned} c_n^{(1)}(t) &= -i \int_0^t dt_1 \, e^{i\omega_{ni} t_1} V_{ni}(t_1) \\ &= -i \int_0^t dt_1 \, e^{i\omega_{ni} t_1} \left(V_{ni} e^{i\omega t_1} + V_{ni}^\dagger e^{-i\omega t_1}\right) \\ &= V_{ni} \frac{1 - e^{i(\omega_{ni} + \omega)t}}{\omega_{ni} + \omega} + V_{ni}^\dagger \frac{1 - e^{i(\omega_{ni} - \omega)t}}{\omega_{ni} - \omega}. \end{aligned}$$

As $t \to \infty$, $\left|c_n^{(1)}\right|^2$ is appreciable only if

$$\omega_{ni} + \omega \approx 0, \qquad \omega_{ni} - \omega \approx 0, \qquad \omega_{ni} = E_n - E_i.$$

Equivalently,

$$E_n \approx E_i - \omega, \qquad E_n \approx E_i + \omega.$$

The corresponding transition rate is

$$\Gamma_{ni} = \frac{dP(i \to n)}{dt} = \begin{cases} 2\pi \overline{|V_{ni}|^2} \rho(E_n), & E_n \approx E_i - \omega, \\ 2\pi \overline{|V_{ni}^\dagger|^2} \rho(E_n), & E_n \approx E_i + \omega. \end{cases} \tag{189}$$

7.7 Rabi Oscillation

Consider a two-level system driven by a sinusoidal perturbation. In the Schrödinger picture,

$$|\Psi(t)\rangle = C_1(t) e^{-iE_1 t}|1\rangle + C_2(t) e^{-iE_2 t}|2\rangle,$$
$$H_0 = E_1|1\rangle\langle 1| + E_2|2\rangle\langle 2|, \qquad V(t) = re^{i\omega t}|1\rangle\langle 2| + re^{-i\omega t}|2\rangle\langle 1|.$$

In the interaction picture, $|\Psi\rangle_I = e^{iH_0 t}|\Psi\rangle$ and $V_I(t) = e^{iH_0 t} V(t) e^{-iH_0 t}$, so that

$$i\frac{d}{dt}\begin{pmatrix} C_1 \\ C_2 \end{pmatrix} = \begin{pmatrix} 0 & re^{i(\omega-\omega_{21})t} \\ re^{-i(\omega-\omega_{21})t} & 0 \end{pmatrix}\begin{pmatrix} C_1 \\ C_2 \end{pmatrix}, \qquad \omega_{21} = E_2 - E_1.$$

For the initial condition $C_2(0) = 0$, $|C_1(0)| = 1$, the exact solution is

$$\begin{aligned} |C_2(t)|^2 &= \frac{r^2}{\Omega^2}\sin^2(\Omega t), \\ |C_1(t)|^2 &= 1 - |C_2(t)|^2, \\ \Omega &= \sqrt{\frac{(\omega-\omega_{21})^2}{4} + r^2}. \end{aligned} \tag{190}$$

At resonance, $\omega = \omega_{21}$, the Rabi frequency reduces to

$$\Omega = r,$$

and the population oscillates completely between $|1\rangle$ and $|2\rangle$:

$$|C_2(t)|^2 = \sin^2(rt), \qquad |C_1(t)|^2 = \cos^2(rt).$$

In contrast to Fermi's golden rule, which predicts a constant transition rate in the weak-coupling limit, the Rabi solution shows that for a discrete two-level system under strong coherent driving the dynamics are fully reversible and periodic.

7.8 Electron transition for H atom

For z-polarized light,

$$\mathbf{E}(t) = E_0 \mathbf{z} \cos(\omega t) = \frac{1}{2} E_0 \mathbf{z}\left(e^{i\omega t} + e^{-i\omega t}\right).$$

The electron-field interaction is

$$V(t) = -e\mathbf{E}(t)\cdot\mathbf{r} = -\frac{1}{2}eE_0 r\cos\theta\left(e^{i\omega t}+e^{-i\omega t}\right).$$

Thus it can be written in the harmonic-perturbation form

$$V(t) = V\left(e^{i\omega t}+e^{-i\omega t}\right), \qquad V = -\frac{1}{2}eE_0 r\cos\theta.$$

The transition matrix element is

$$\begin{aligned} V_{fi} &= \langle f|V|i\rangle = -\frac{1}{2}eE_0\langle f|r\cos\theta|i\rangle \\ &= -\frac{1}{2}eE_0\int \psi_f^*\psi_i\,(r\cos\theta)\,dV \\ &= -\frac{1}{2}eE_0\int R_f^*Y_f^*\,R_iY_i\,r\cos\theta\,r^2\sin\theta\,dr\,d\theta\,d\phi \\ &= -\frac{1}{2}eE_0\left[\int_0^\infty R_f^*(r)R_i(r)\,r^3dr\right] \\ &\quad\times\left[\int Y_f^*(\theta,\phi)Y_i(\theta,\phi)\sin\theta\cos\theta\,d\theta\,d\phi\right]. \end{aligned} \tag{191}$$

7.9 Selection Rules for Electric Dipole (E1) Transitions

From the matrix element derived above, the angular integral

$$\int Y_f^*(\theta,\phi)Y_i(\theta,\phi)\sin\theta\cos\theta\,d\theta\,d\phi$$

determines whether V_{fi} vanishes. Using

$$\cos\theta \propto Y_{1,0}(\theta,\phi)$$

and the Wigner-Eckart theorem, one obtains the electric-dipole selection rules:

$$\begin{cases} \text{Parity: } \pi_f = -\pi_i, \\ \text{Orbital angular momentum: } \Delta\ell = \pm 1, \\ \text{Magnetic quantum number: } \Delta m = 0,\ \pm 1. \end{cases} \tag{192}$$

For multi-electron atoms, these generalize to

$$\begin{cases} \text{Parity: } \pi_f = -\pi_i, \\ \text{Total angular momentum: } \Delta J = 0,\ \pm 1 \quad (J_i = 0 \to J_f = 0 \text{ forbidden}), \\ \Delta m_J = 0,\ \pm 1, \\ \text{Spin: } \Delta S = 0, \\ \text{Orbital: } \Delta L = 0,\ \pm 1 \quad (L_i = 0 \to L_f = 0 \text{ forbidden}). \end{cases} \tag{193}$$

These rules follow directly from Fermi's golden rule and the electric-dipole interaction Hamiltonian

$$H_{\text{int}} = -e\mathbf{E}\cdot\mathbf{r},$$

where $e > 0$ is the elementary charge.